

Technical Report

Introduction

This project analyzes customer churn and sales data.

Data Cleaning

- Removed duplicates
- Handled missing values
- Standardized dataset

Exploratory Data Analysis

- Churn distribution analyzed
- Sales trends visualized
- Correlation heatmap created

Statistical Analysis

Hypothesis testing performed using T-test.

Machine Learning

Random Forest Classifier used for churn prediction.

Results

Model achieved good prediction accuracy.

Recommendations

- Retention strategies
- Marketing optimization
- Customer segmentation targeting

Conclusion

Data-driven decisions can significantly improve business performance.